

L05 Data Preprocessing Generative AI

Lab thoughts / Reflective Journal

What I learned in this lab:

Completing this lab further increased my understanding of how important preprocessing is. Far beyond just cleaning input data before applying ML models or other transformations, preprocessing decisions fundamentally impact the quality of every step down the pipeline.

I have worked with text data before but often got caught up in downstream tasks and made assumptions about preprocessing. It was helpful to work through this lab step by step to see the decisions I make when preprocessing data impact my results later in the machine learning pipeline. The lab also took longer to troubleshoot than I expected and I am glad I allocated enough time to complete it and this journal.

Challenges

Some challenges I faced while working through this lab include when generating embeddings for BERT. The notebook got hung for upwards of twenty minutes while generating sentence embeddings. I had to go back and try to troubleshoot what was happening behind the scenes. After digging deeper, I realized that I had accidentally recomputed embeddings and was running all computations on CPU. I updated my code to use GPU and removed unnecessary code chunks and was immediately rewarded with results.

Another challenge occurred early in the lab when first computing similarity statistics for same-category pairs. My output was printing NaN because I didn't have enough same-category pairs initially to compare. Strategically sampling with stratify was something I didn't even consider preprocessing at first but was vital to completing comparison later in the lab.

What I Learned about Data Preprocessing

This lab taught me that there is no golden rule for preprocessing data. Simple changes like removing stopwords or how aggressive I want to be with normalizing text changed how each embedding type behaved.

BERT was much more forgiving of preprocessing choices as it is designed to understand contextual information. Removing stopwords and normalizing sentence structure had less impact on BERT but dramatically changed how well GloVe could separate categories.

I also learned that preprocessing isn't just about cleaning text data but preparing data specifically for the algorithms I want to use. GloVe could not distinguish between categories as well as BERT because every word has a fixed meaning. However, thanks to BERT's transformer architecture, it produced significantly better category separation.

Understanding this became very clear when looking at the distributions of similarities between categories, results of statistical hypothesis testing, and overall accuracy scores.

Visualization also helped me better understand my results. Generating histograms and boxplots allowed me to quickly see how each embedding performed relative to the others. Viewing these side by side made it very obvious why BERT outperformed GloVe and how preprocessing can affect these models.

How This Applies to Generative AI

The same concepts around embeddings and preprocessing applied to generative AI. High-quality embeddings and well-structured input data are vital to the success of any generative AI model.

Generative AI can do text summarization, translation, conversation, content moderation, or even retrieval-augmented generation. However, it all starts with understanding. The model can only generate intelligent outputs if it has access to high-quality data.

Techniques from this lab like cleaning input data, normalizing structure, and making sure sampling between categories was balanced are necessary for successful generative AI. Here are some real-world applications of text generation and where preprocessing plays a key role:

Needs clean inputs which prompt engineering can help facilitate.

Creating datasets to fine-tune generative models require preprocessing to ensure not amplifying noise.

Systems that utilize embedding-based retrieval would need strong separation (i.e., what BERT provided in this lab).

Working through this lab made me appreciate how preprocessing isn't just a nice-to-have beginning step but rather an important design decision.

What surprised me

I was surprised by how high the same-category similarities were for both GloVe and BERT. It made me realize just how clean the BBC dataset was. I was surprised by how poorly the GloVe pretrained embeddings could separate the different categories. While I knew BERT would perform better, I did not realize there would be such a drastic difference.

Real-World Uses

Some real-world applications of preprocessing text data include:

- News classification/topic modeling
- Search/recommendation engines
- Moderation/clustering
- Retrieval-augmented generation
- Sentiment/opinion analysis

If you need to understand text at a semantic level, preprocessing your data and creating high-quality embeddings is necessary.

Areas I Want to Focus On

Areas I want to focus on in the future include learning about different preprocessing techniques. If possible, I want to look into how some of the large generative AI systems preprocess data. Some ideas include:

- Playing with domain-specific models like BioBERT or FinBERT.
- Bias detection/mitigation.
- Chunking/hierarchical embeddings with longer documents.
- Building rudimentary retrieval-augmented generation pipelines to see how preprocessing impacts retrieval quality.